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Brain asymmetries of attention and consciousness

Paolo Bartolomeo

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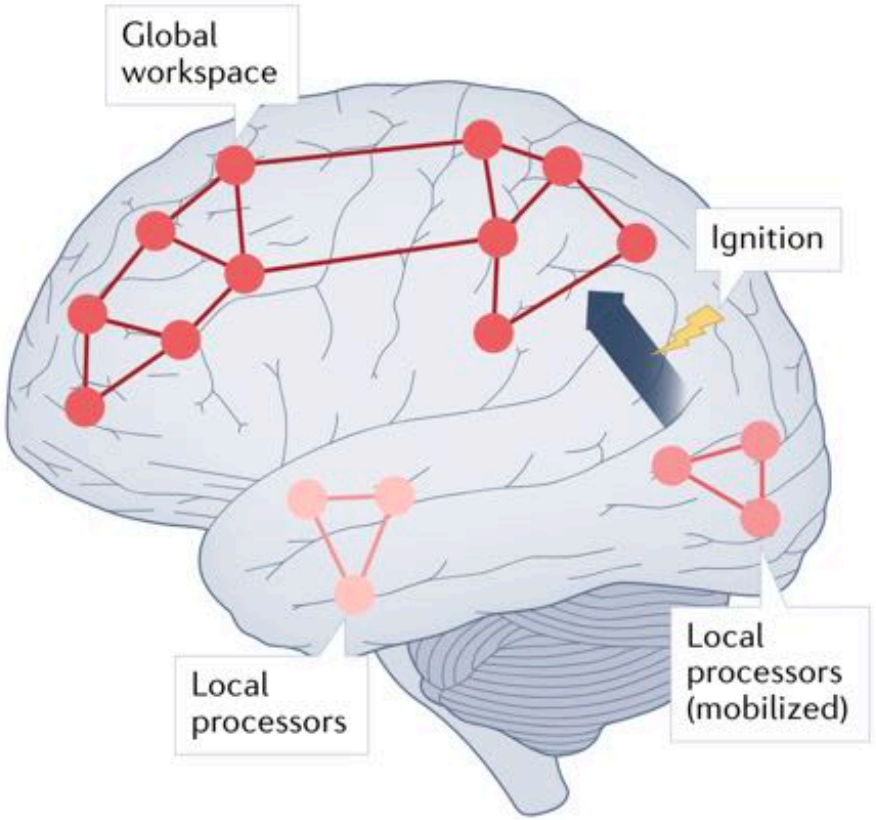
22 theories of consciousness

Table 1 | **A selection of theories of consciousness**

Theory	Primary claim	Key refs
Higher-order theory (HOT)	Consciousness depends on meta-representations of lower-order mental states	[illegible]
Self-organizing meta-representational theory	Consciousness is the brain's (meta-representational) theory about itself	[illegible]
Attended intermediate representation theory	Consciousness depends on the attentional amplification of intermediate-level representations	[illegible]
Global workspace theories (GWTs)	Consciousness depends on ignition and broadcast within a neuronal global workspace where fronto-parietal cortical regions play a central, hub-like role	[illegible]
Integrated information theory (IIT)	Consciousness is identical to the cause–effect structure of a physical substrate that specifies a maximum of irreducible integrated information	[illegible]
Information closure theory	Consciousness depends on non-trivial information closure with respect to an environment at particular coarse-grained scales	[illegible]
Dynamic core theory	Consciousness depends on a functional cluster of neural activity combining high levels of dynamical integration and differentiation	[illegible]
Neural Darwinism	Consciousness depends on re-entrant interactions reflecting a history of value-dependent learning events shaped by selectionist principles	[illegible]
Local recurrency	Consciousness depends on local recurrent or re-entrant cortical processing and promotes learning	[illegible]
Predictive processing	Perception depends on predictive inference of the causes of sensory signals; provides a framework for systematically mapping neural mechanisms to aspects of consciousness	[illegible]
Neuro-representationalism	Consciousness depends on multilevel neurally encoded predictive representations	[illegible]
Active inference	Although views vary, in one version consciousness depends on temporally and counterfactually deep inference about self-generated actions	[illegible]; see also[illegible]
Beast machine theory	Consciousness is grounded in allostatic control-oriented predictive inference	[illegible]; see also[illegible]
Neural subjective frame	Consciousness depends on neural maps of the bodily state providing a first-person perspective	[illegible]
Self comes to mind theory	Consciousness depends on interactions between homeostatic routines and multilevel interoceptive maps, with affect and feeling at the core	[illegible]
Attention schema theory	Consciousness depends on a neurally encoded model of the control of attention	[illegible]
Multiple drafts model	Consciousness depends on multiple (potentially inconsistent) representations rather than a single, unified representation that is available to a central system	[illegible]
Sensorimotor theory	Consciousness depends on mastery of the laws governing sensorimotor contingencies	[illegible]
Unlimited associative learning	Consciousness depends on a form of learning which enables an organism to link motivational value with stimuli or actions that are novel, compound and non-reflex inducing	[illegible]
Dendritic integration theory	Consciousness depends on integration of top-down and bottom-up signalling at a cellular level	[illegible]
Electromagnetic field theory	Consciousness is identical to physically integrated, and causally active, information encoded in the brain's global electromagnetic field	[illegible]
Orchestrated objective reduction	Consciousness depends on quantum computations within microtubules inside neurons	[illegible]

Our selection of theories includes those that are either neurobiological in nature or potentially expressible in neurobiological terms.



Seth & Bayne *Nat Rev Neurosci* **23**:439, 2022

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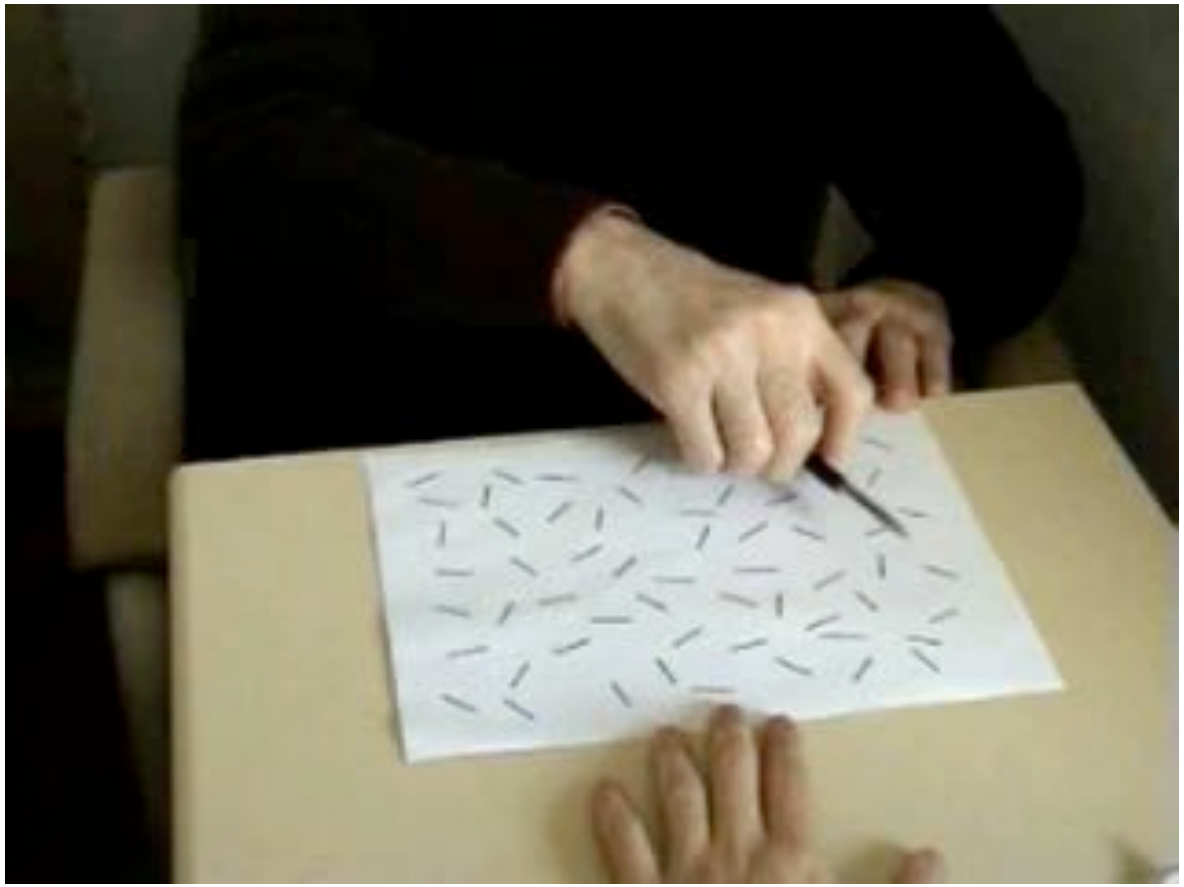
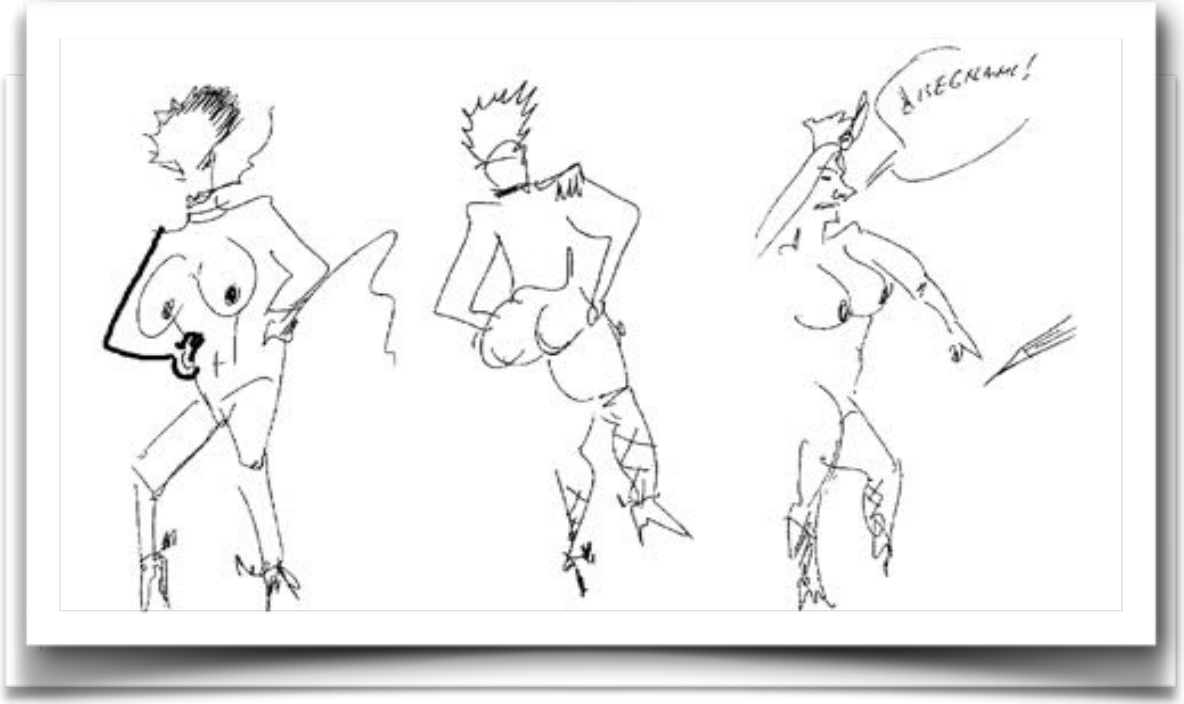
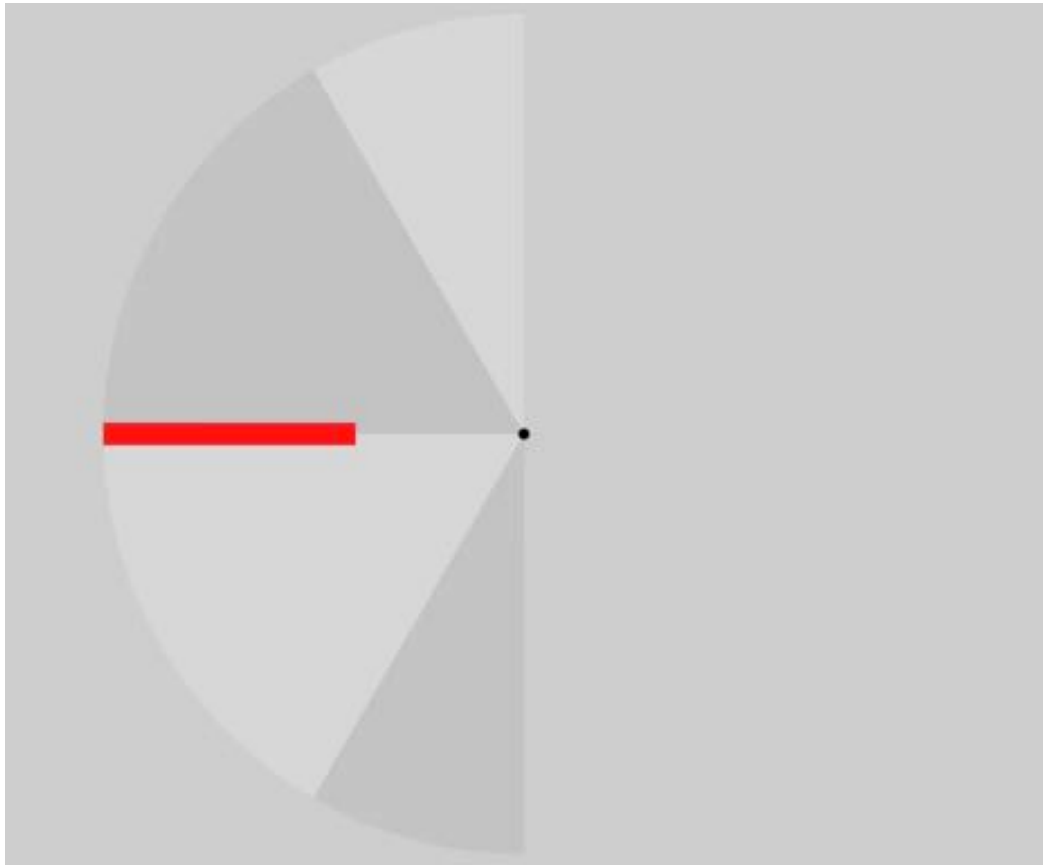
Spatial neglect



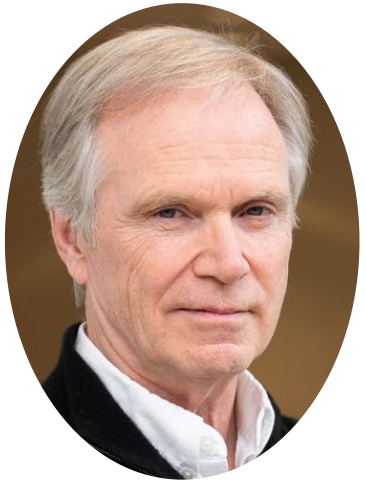


Cantagallo & Della Sala, *Cortex* **34**:163, 1998

Spatial neglect: When brain damage “improves” perception



Stefania de Vito



Patrick Cavanagh

de Vito… Cavanagh & Bartolomeo *J Neurophysiol* **114**:3351, 2015

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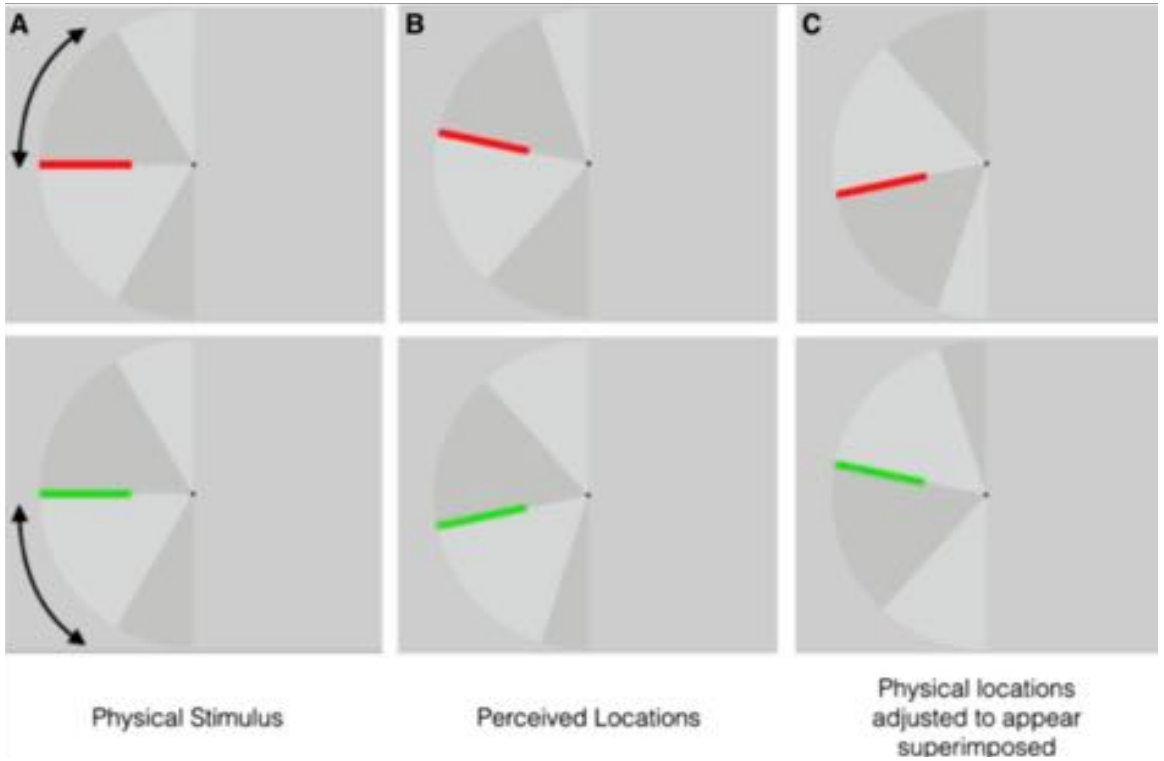

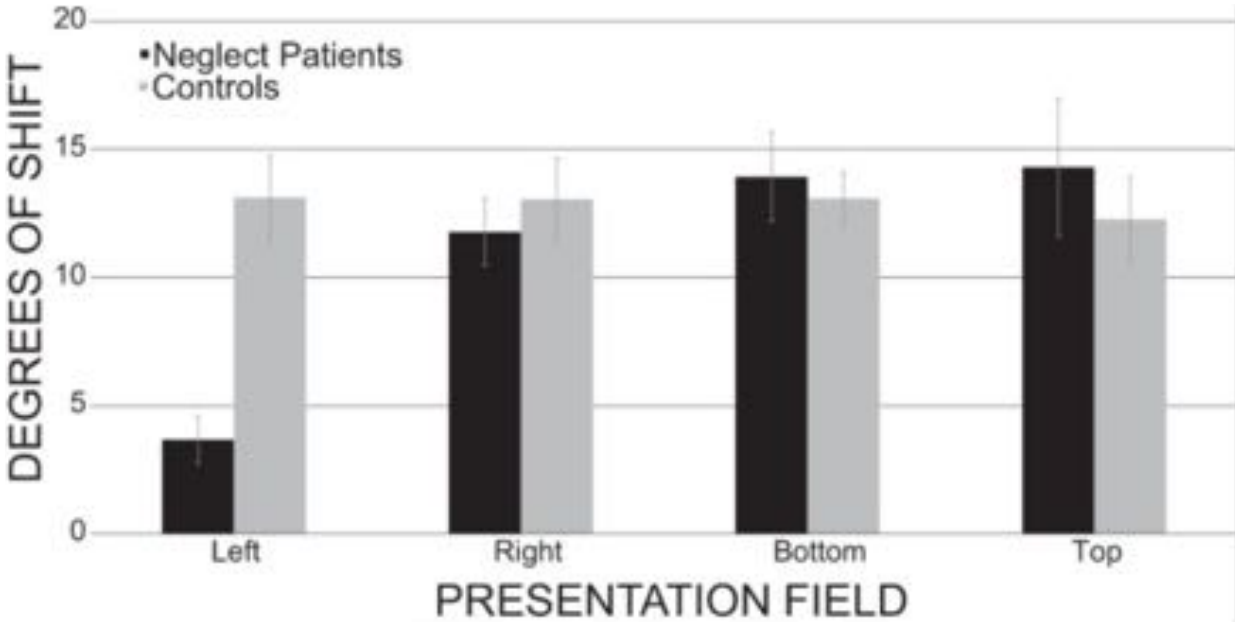

de Vito… Cavanagh & Bartolomeo *J Neurophysiol* **114**:3351, 2015

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Direct Evidence for a Parietal-Frontal Pathway Subserving Spatial Awareness in Humans

Michel Thiebaut de Schotten,[1] Marika Urbanski,[1] Hugues Duffau,[2]
Emmanuelle Volle,[1,3] Richard Lévy,[1,4] Bruno Dubois,[1,4]
Paolo Bartolomeo[1,4*]

Science **309**:2226, 2005

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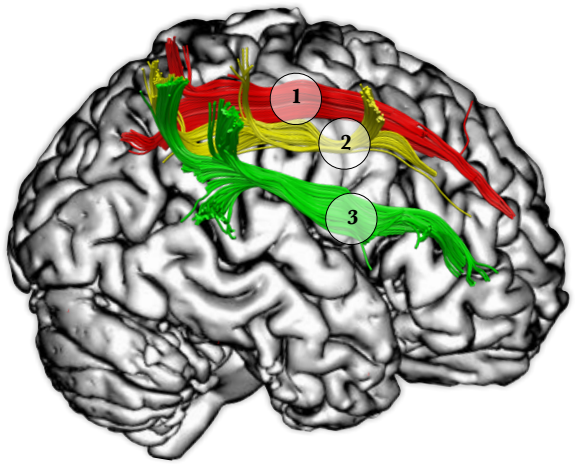
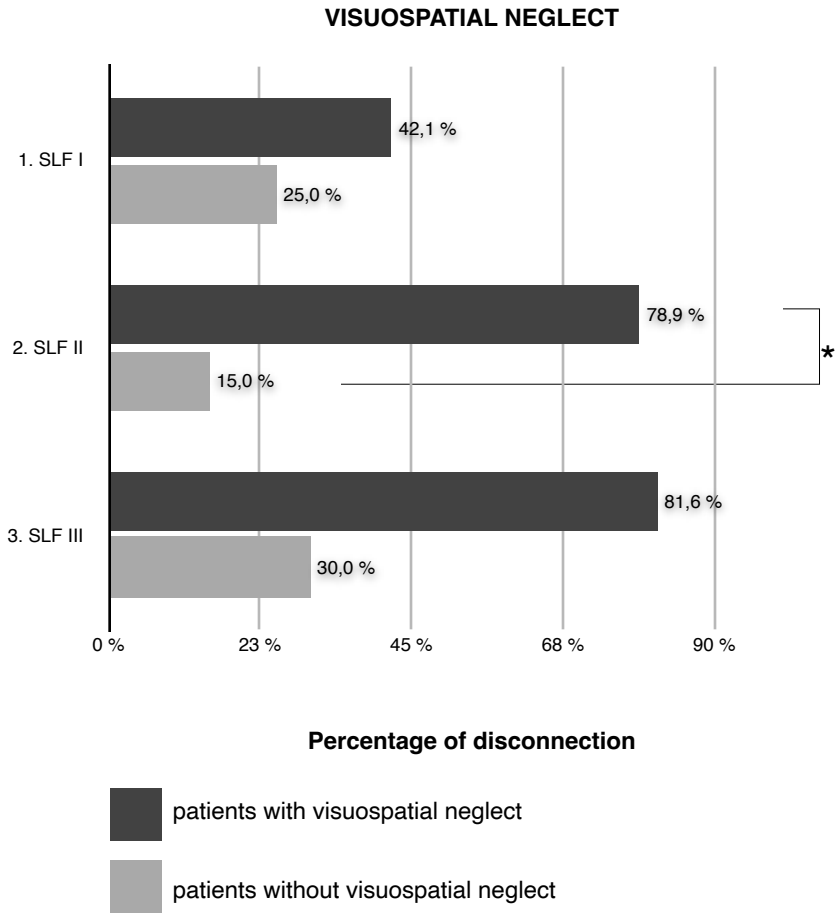
Spatial neglect





58 patients

Thiebaut de Schotten… Bartolomeo & Doricchi, *Cerebral Cortex* **24**:691, 2014

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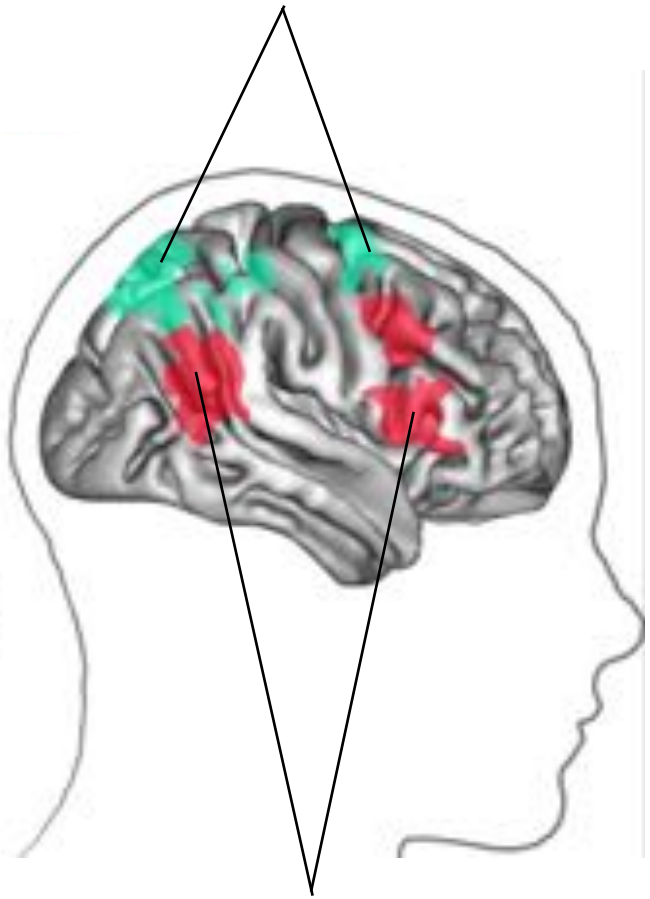
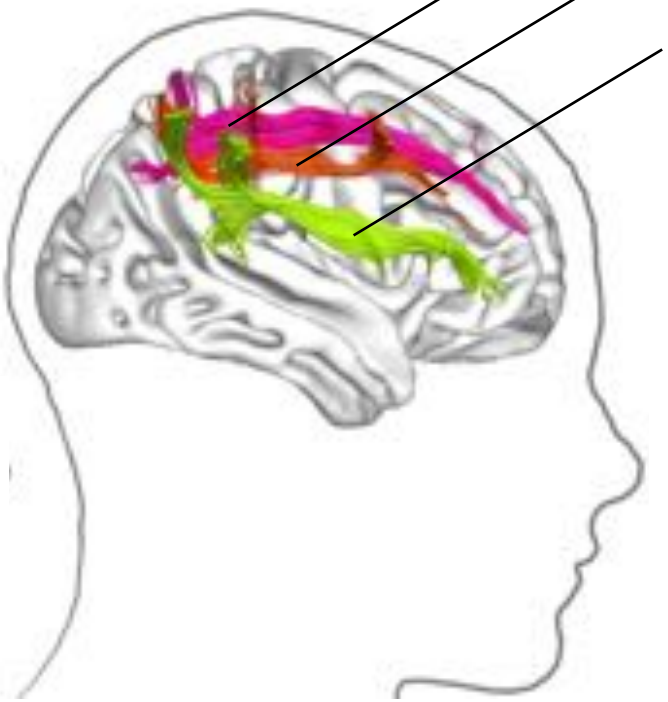
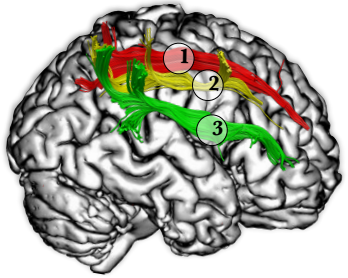




Corbetta & Shulman, *Nat Rev Neurosci* **3**:201-15, 2002

Thiebaut de Schotten et al, *Nat Neurosci* **14**:1245-6, 2011

• Interim conclusion



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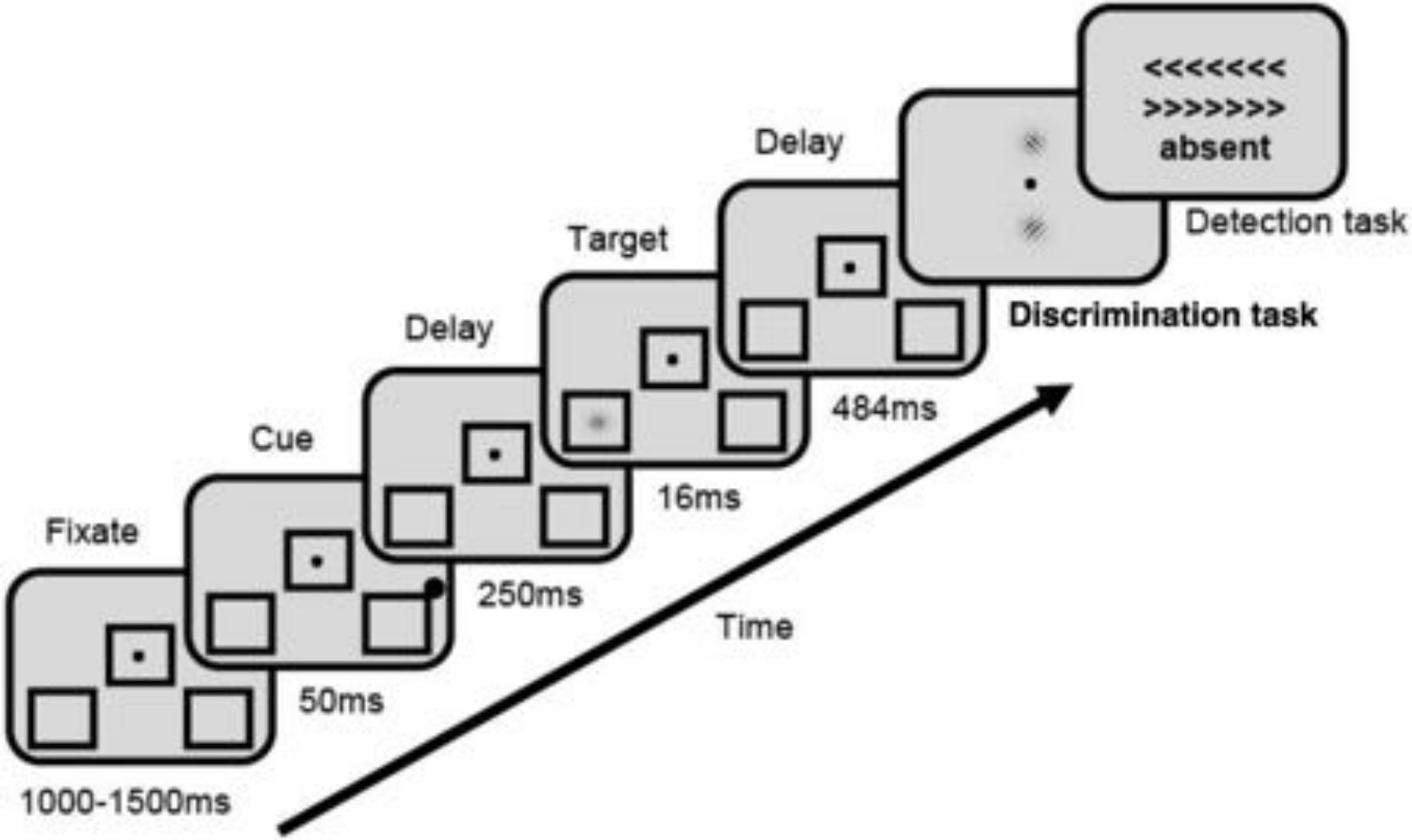
Posner-type task



Ana B. Chica

Chica et al. *NeuroImage* 2010, *Atten Percept Psychophys* 2011, *Cereb Cortex* 2013, 2014

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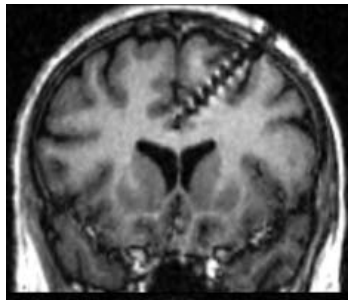
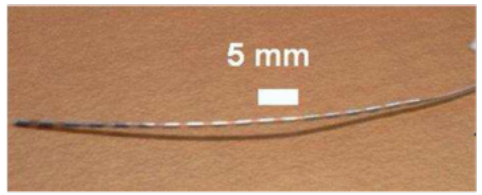
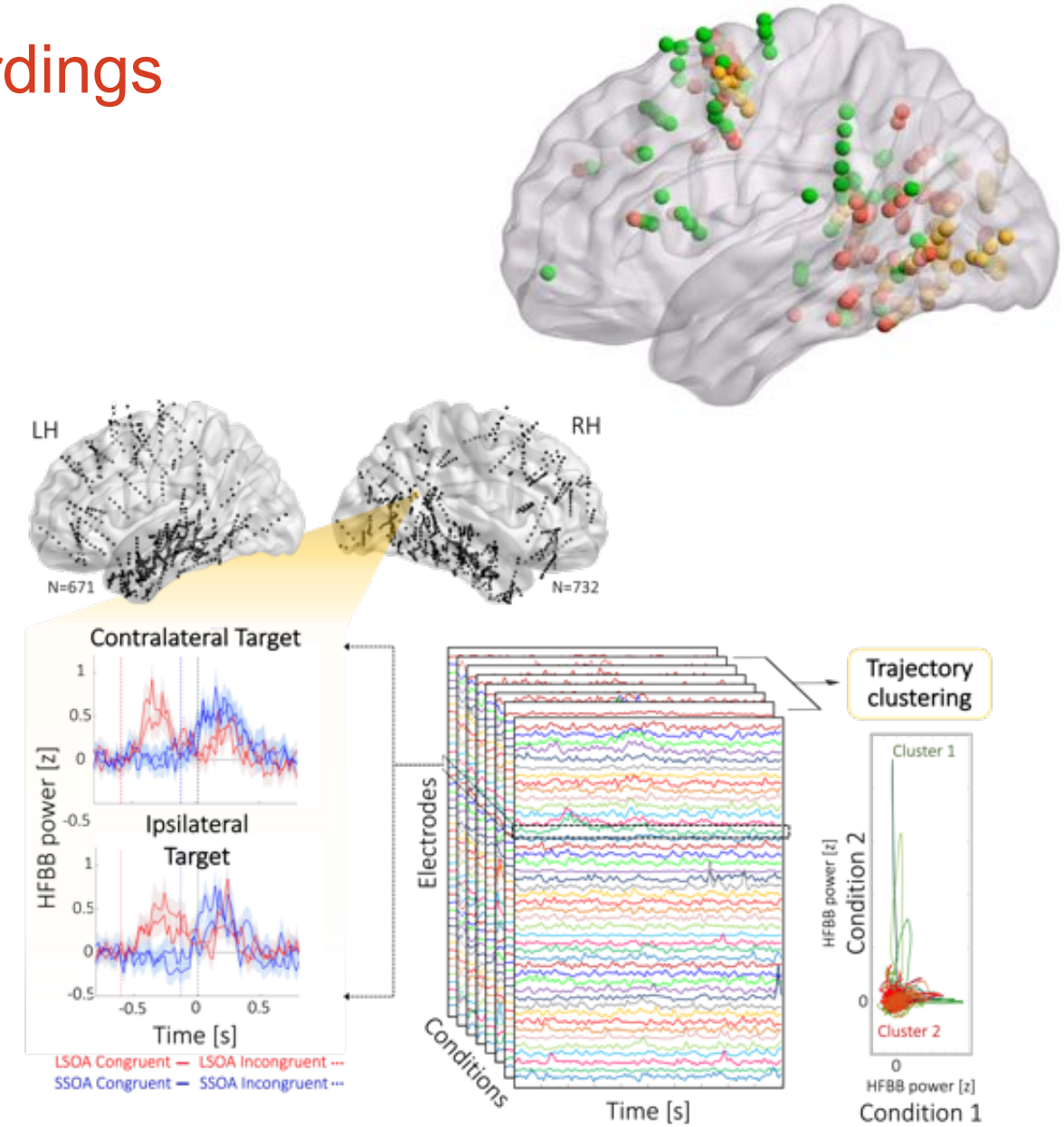
Intracerebral recordings







Tal Seidel Malkinson

Seidel Malkinson… & Bartolomeo *Nature Communications* **15**:2586, 2024

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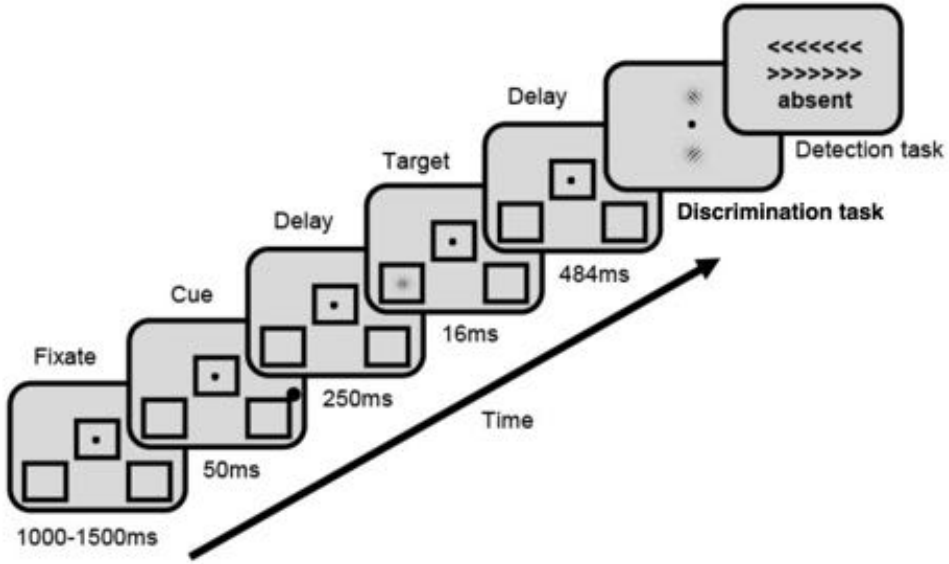
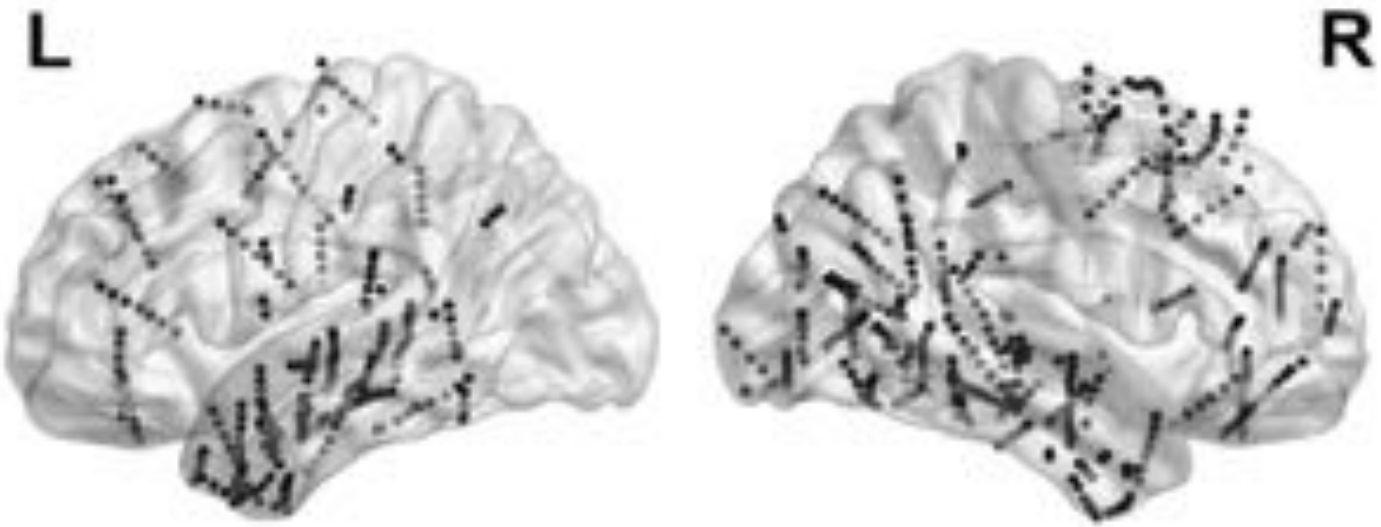
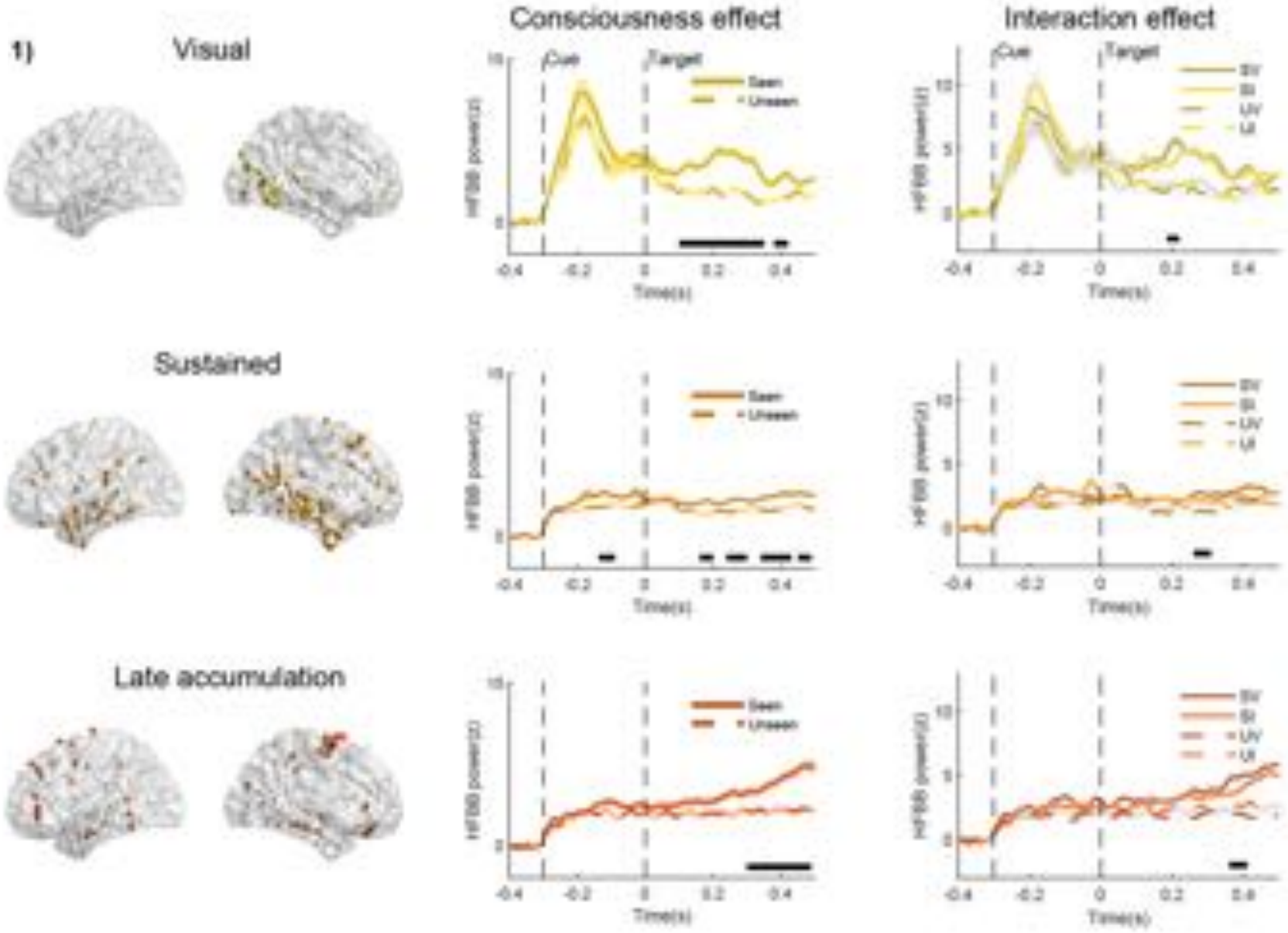




Jianghao Liu

727 intracerebral contacts in 13 epilepsy patients

Liu… Seidel Malkinson & Bartolomeo *Commun Biol* **6**:730, 2023



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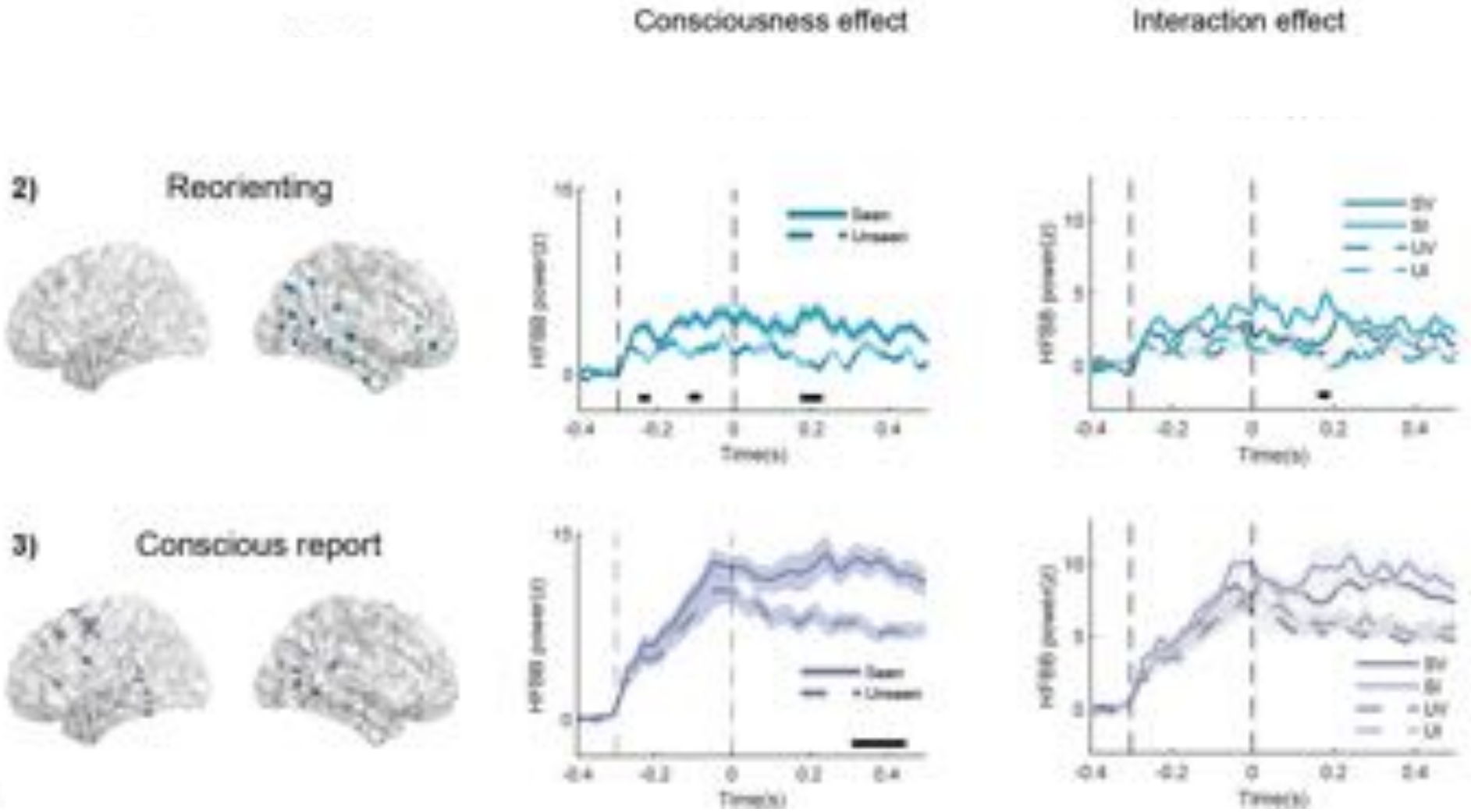

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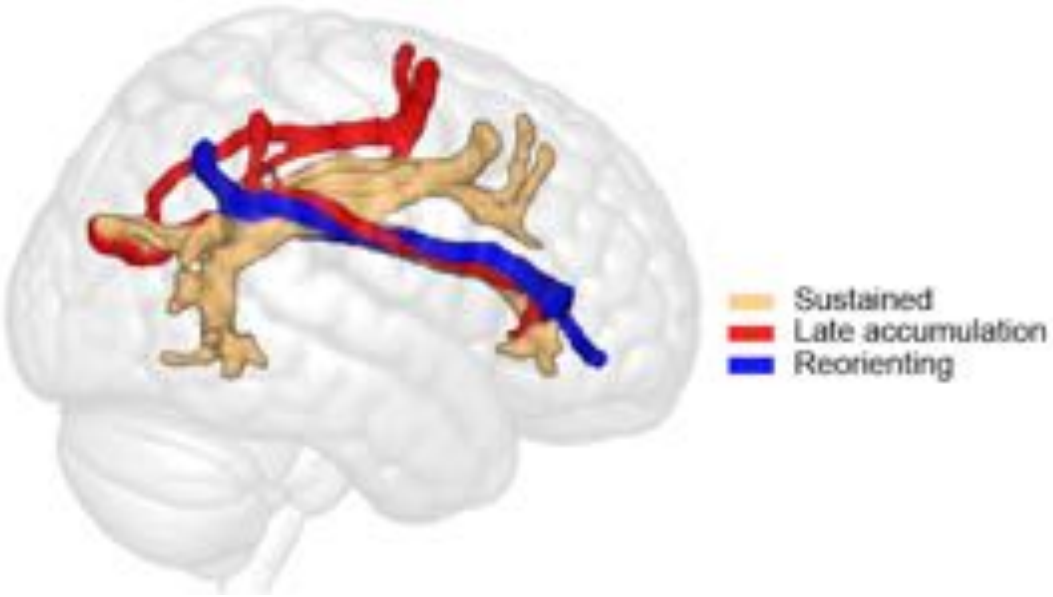
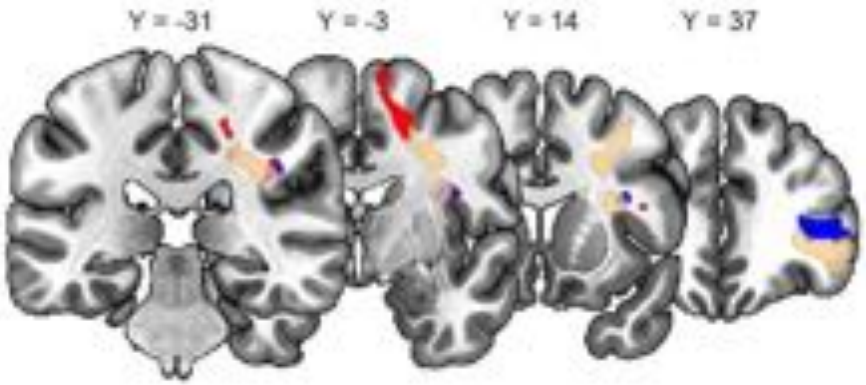
WM tractography





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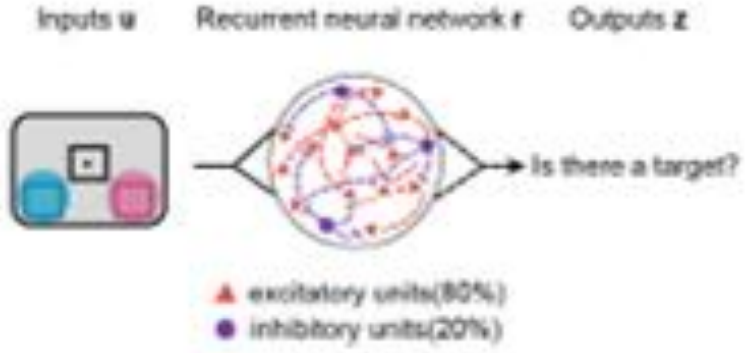
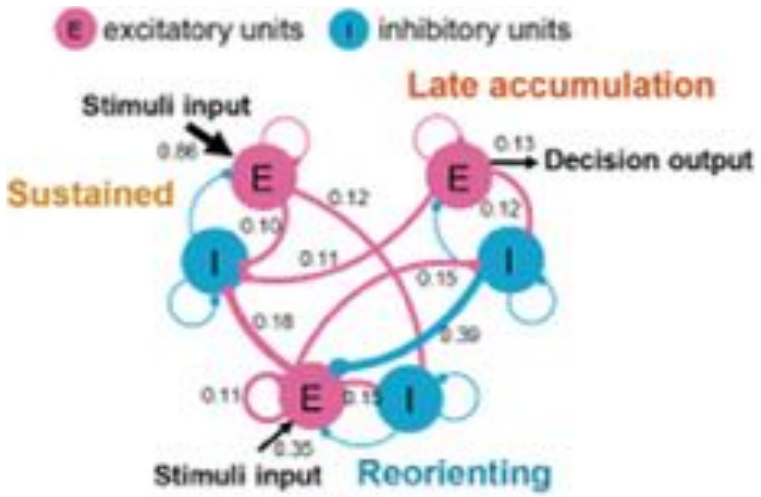
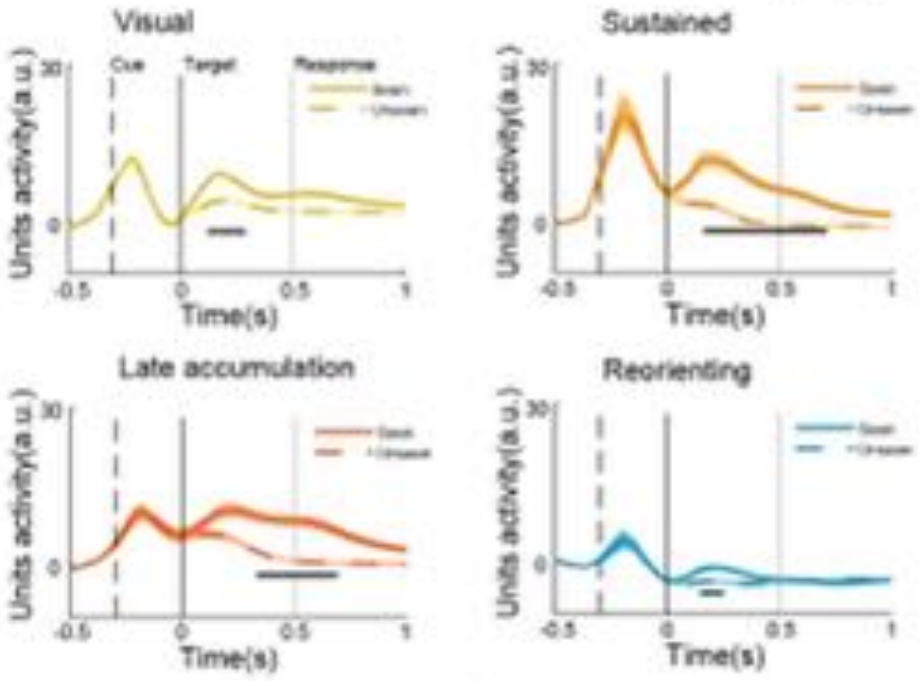
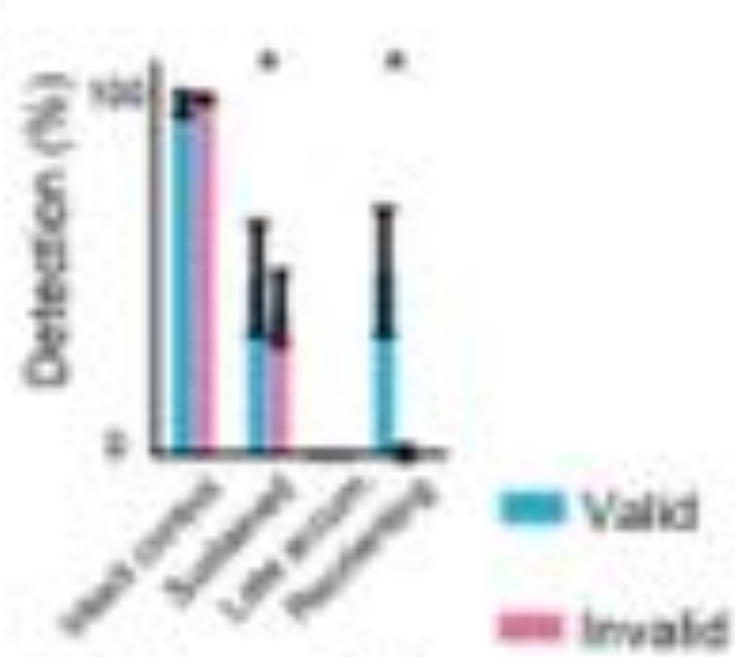
Neural network model









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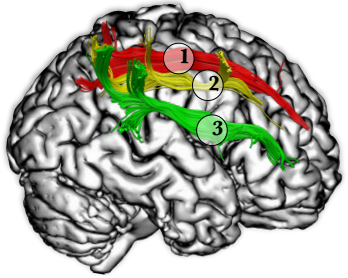
• Interim conclusions



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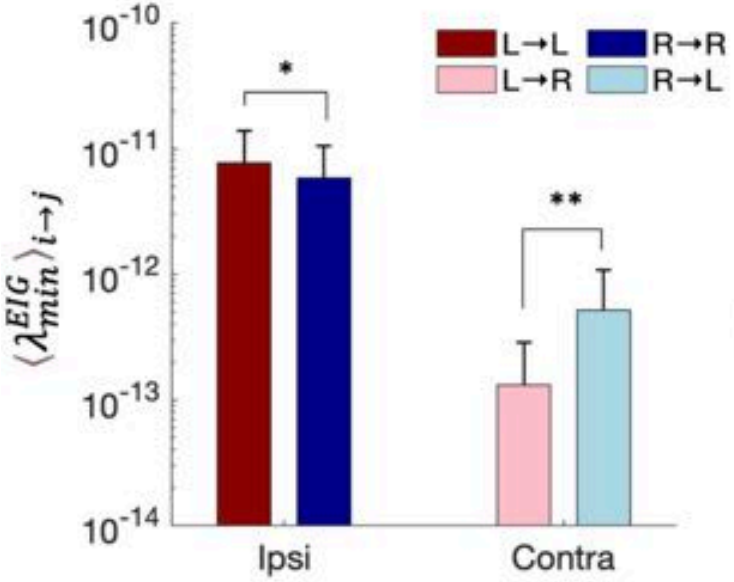
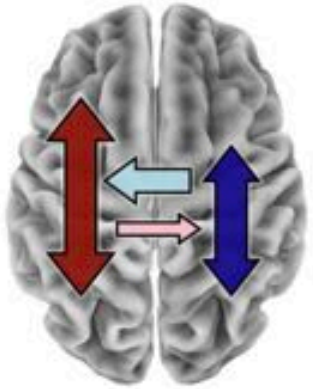
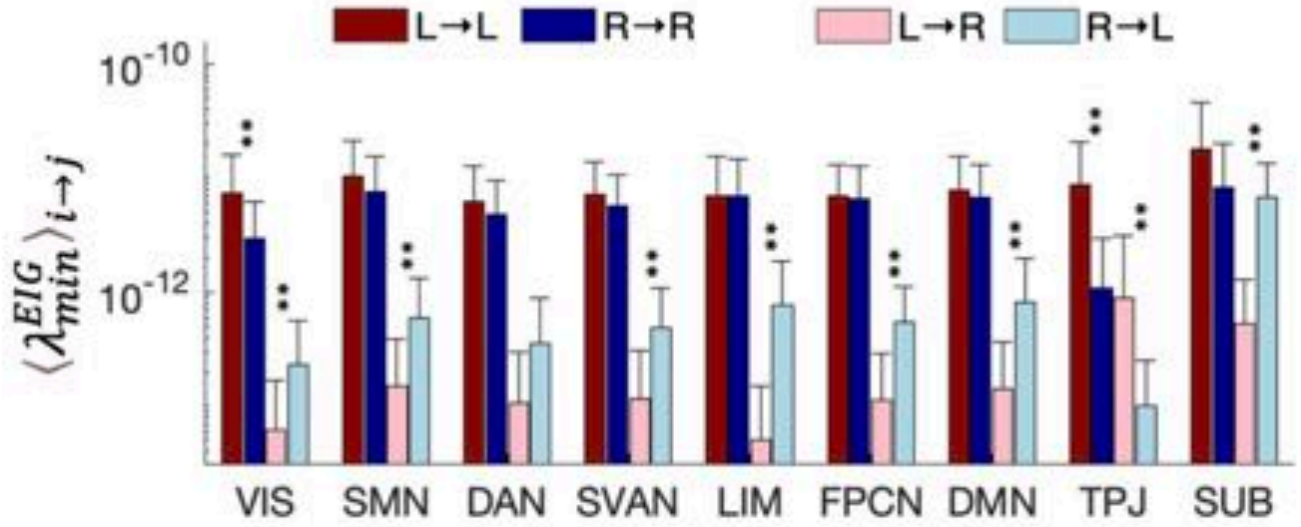
2. Distinct roles of RH SLF networks in visual awareness: attention maintenance, evidence accumulation, reorienting

What's so special with the RH?



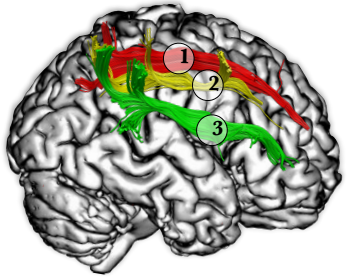


Fabrizio de Vico Fallani



Ben Messaoud… & De Vico Fallani. Low-dimensional controllability of brain networks. *PLoS Comput Biol* **21**:e1012691, 2025

• Interim conclusions



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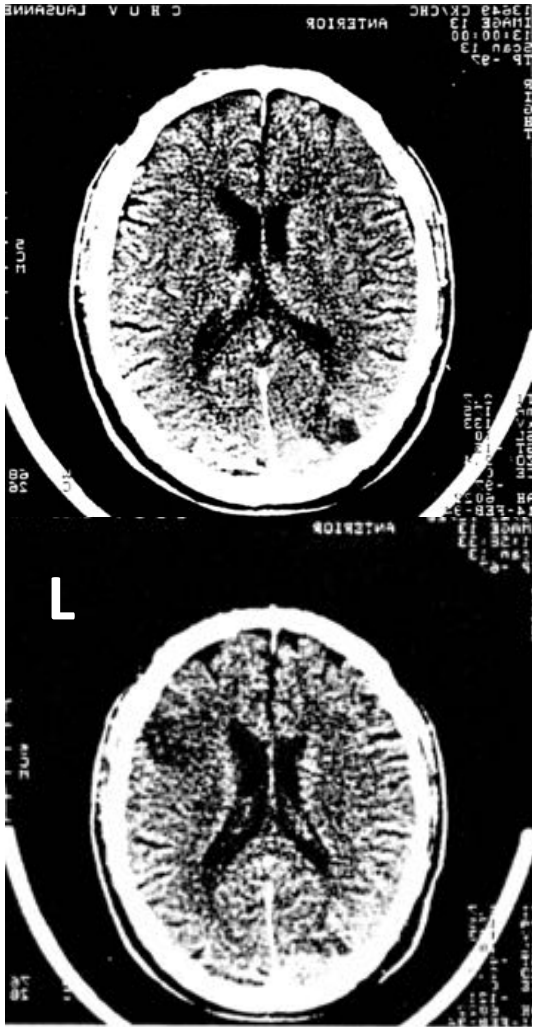
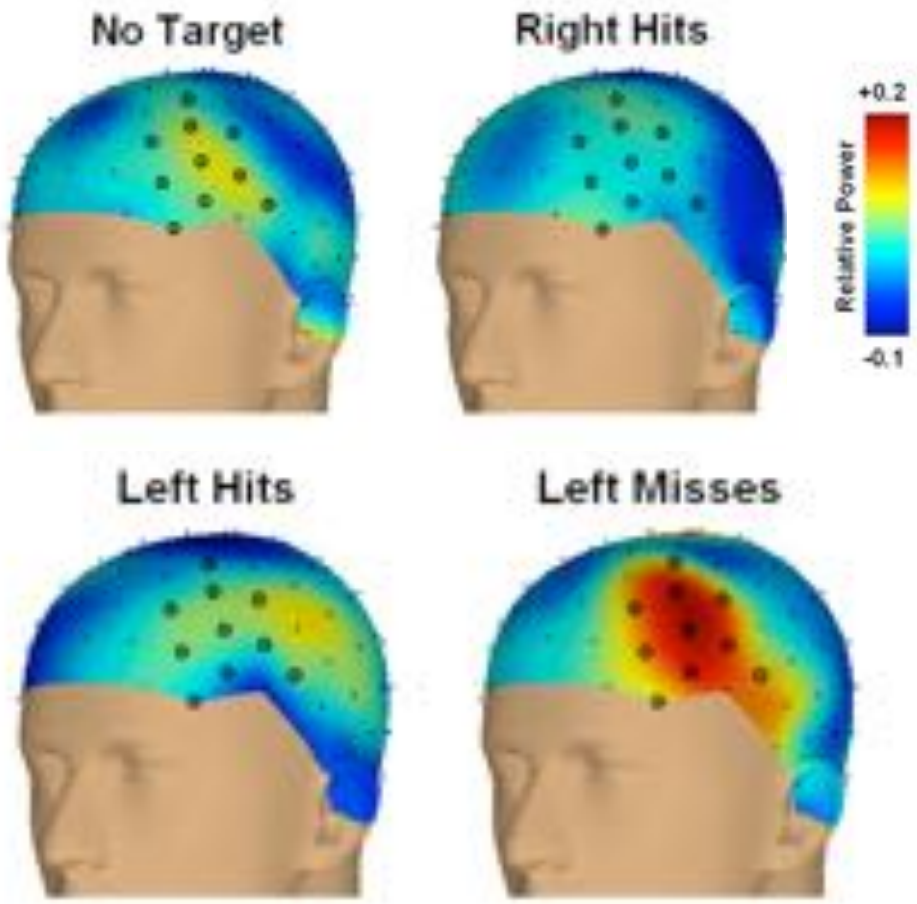
What about the LH PFC?

LH PFC and conscious report: Back to neglect



Vuilleumier et al. *Neurology* **46**:184, 1996



Rastelli... & Bartolomeo, *Cortex* **49**:1989, 2013

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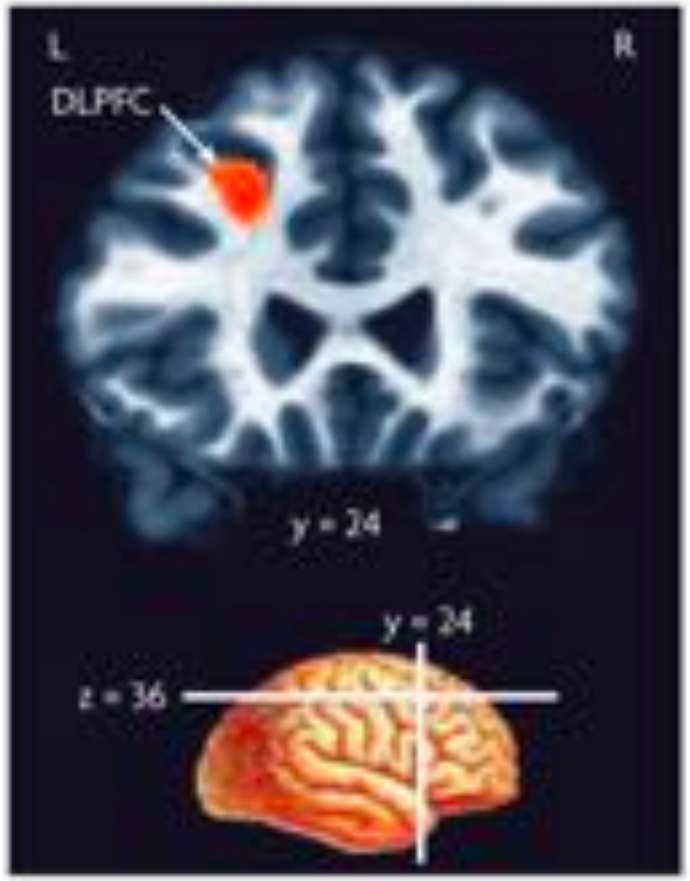
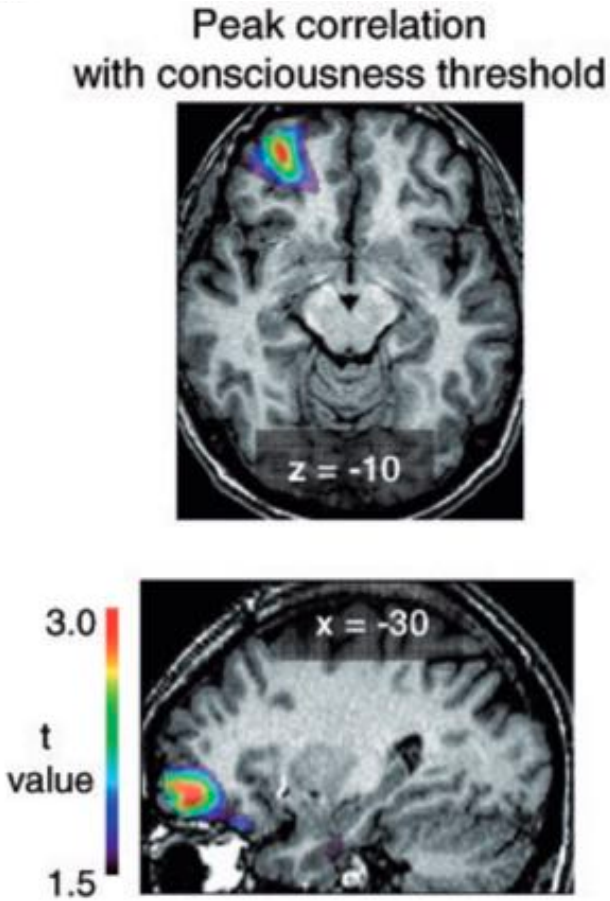
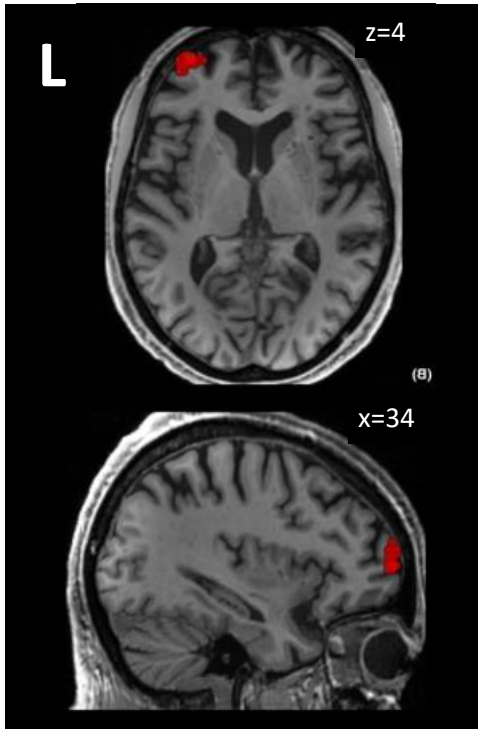
LH PFC and conscious report: Visual perception



Heekeren… & Ungerleider, *Nature* **431**:859, 2004

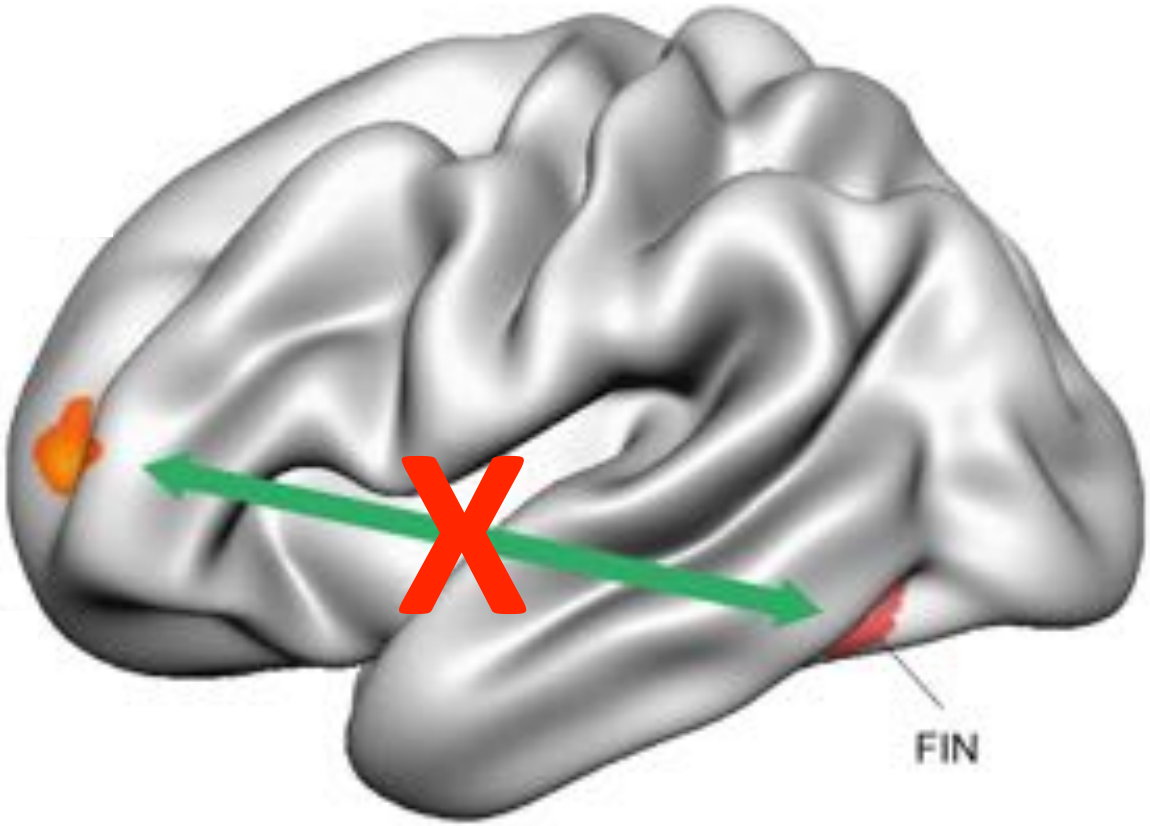


Del Cul et al, *Brain* **132**:2531, 2009



Hauw… & Naccache, Are we aware of neural activity in primary visual cortex? A neuropsychological case study. *Ann Clin Trans Neurol* **11**:1365, 2024

LH PFC and subjective experience: Aphantasia



Alfredo Spagna

Jianghao Liu

Liu… & Bartolomeo *Cortex*, **185**: 113, 2025

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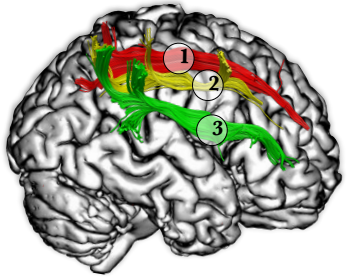
• Interim conclusions



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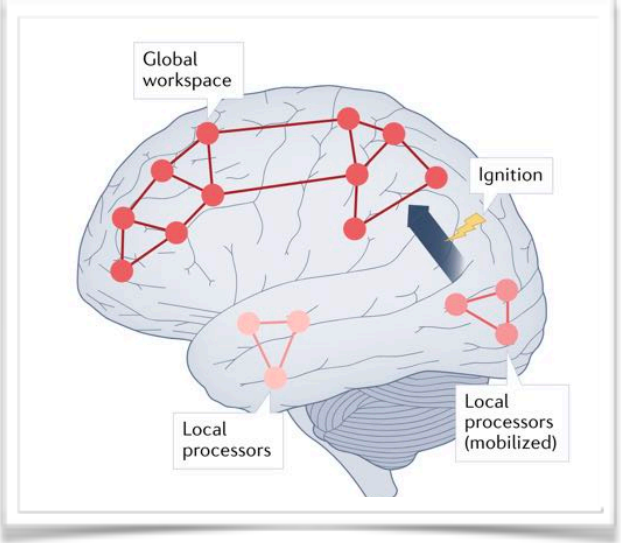
2. Distinct roles of RH SLF networks in visual awareness: attention maintenance, evidence accumulation, reorienting

3. Possible roles of LH PFC in conscious report: perceptual decisions, model building?

• General conclusion

Coordinated but distinct roles of bilateral F-P networks in visual awareness: specifying the GNW hypothesis



CellPress

Trends in Cognitive Sciences

Frontoparietal asymmetries leading to conscious perception

Paolo Bartolomeo [1,*,@], Jianghao Liu [1,2], and Tal Seidel Malkinson [3]

Trends in Cognitive Sciences **29**:222, 2025

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